

Project Charter

UC2 - Enterprise Healthcare Data Platform (EHDP)

1. Project Information

Project Name	Enterprise Healthcare Data Platform (EHDP)
Use Case	UC2 – Enterprise Healthcare Data Platform
Domain	Healthcare
Project Methodology	Agile Scrum
Project Duration	3 Months
Primary ETL Tool	Pentaho Data Integration
Database	PostgreSQL
Programming / Data Processing	Python (Pandas)
Reporting Tool	Power BI
Version Control	Git
Data Source	Synthea Synthetic Healthcare Dataset

2. Project Background

ABC Healthcare Group operates a network of 12 multi-specialty hospitals and 35 diagnostic centers across India. Healthcare information is generated from multiple systems, including Hospital Information Systems, Electronic Health Records, Laboratory Information Systems, Pharmacy, Billing, Insurance, Radiology, ICU monitoring devices, and Appointment Management Systems.

Because these data sources are fragmented and disconnected, hospital administrators face difficulties in monitoring patient flow, optimizing resource utilization, tracking clinical outcomes, and improving operational efficiency.

The project aims to develop an Enterprise Healthcare Data Platform that integrates healthcare data from multiple sources and provides trusted data for hospital analytics, operational reporting, and executive decision-making.

3. Problem Statement

Healthcare data is generated in multiple formats and from multiple healthcare-related systems. The fragmented nature of these sources makes it difficult to obtain a unified and reliable view of healthcare information.

The project therefore requires an enterprise data platform capable of:

- Ingesting data from heterogeneous healthcare sources.
- Cleaning, validating, and standardizing healthcare data.
- Storing data in PostgreSQL.
- Maintaining metadata and data lineage.
- Providing trusted datasets for analytics and reporting.

For this implementation, the **Synthea Synthetic Healthcare Dataset** is used as the healthcare data source. The available datasets include patient, condition, observation, medication, procedure, allergy, immunization, provider, payer, organization, and other healthcare-related data.

4. Project Objective

The main objective is to design and implement an enterprise-grade healthcare data platform that:

1. Collects healthcare data from multiple heterogeneous sources.
 2. Integrates the Synthea healthcare datasets into the platform.
 3. Cleans, validates, and standardizes healthcare data.
 4. Uses Pentaho Data Integration to develop ETL pipelines.
 5. Stores curated datasets in an enterprise PostgreSQL Data Warehouse.
 6. Maintains metadata, audit information, and data lineage as the project progresses.
 7. Supports hospital operations, patient care analytics, and executive reporting.
 8. Uses Git and GitHub for collaborative version control.
 9. Produces technical documentation and sprint deliverables.
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5. Data Source

Synthea Synthetic Healthcare Dataset

The project uses Synthea as the healthcare data source.

Website: <https://synthetichealth.github.io/synthea/#open-source-landing>

The available datasets include:

- Patients
- Encounters
- Conditions
- Observations
- Medications

- Procedures
- Immunizations
- Allergies
- Careplans
- Devices
- Imaging Studies
- Organizations
- Providers
- Payers
- Payer Transitions
- Supplies

The datasets are available in different file formats, CSV and JSON, providing an opportunity to demonstrate heterogeneous healthcare data ingestion using Pentaho.

6. Key Stakeholders

The UC2 project identifies the following major healthcare stakeholders:

Stakeholder	Interest / Responsibility
Doctors	Access to reliable clinical information
Hospital Administrators	Monitoring hospital operations and patient flow
Finance Team	Healthcare billing and financial information
Operations Team	Resource utilization and operational efficiency
Management	Executive-level analytics and decision-making
Data Engineering Team	Development and maintenance of the healthcare data platform
Data Engineering Lead	Project guidance and technical direction

7. Project Scope

In Scope

The project will include:

- Analysis of healthcare data sources.

- Synthea dataset integration.
- Healthcare source inventory.
- Data dictionary preparation.
- Pentaho ETL pipeline development.
- Ingestion of healthcare data from different file formats.
- Data validation and transformation.
- PostgreSQL staging database.
- ETL logging and exception handling.
- Git/GitHub version control.
- Source-to-target documentation.
- Data quality documentation.
- Sprint review presentations.

Out of Scope for Sprint 0 and Sprint 1

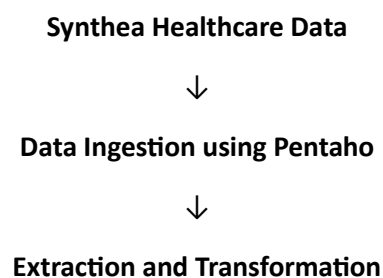
The following activities are planned for later sprints rather than being the primary focus of Sprint 0 and Sprint 1:

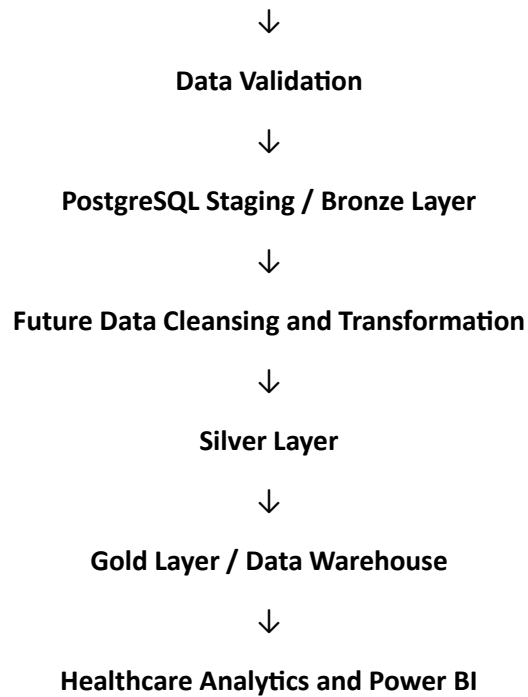
- Detailed Python data profiling.
- Complete data quality assessment.
- Star schema design.
- Fact and dimension tables.
- Full PostgreSQL data warehouse development.
- Advanced data lineage implementation.
- Pentaho ETL orchestration jobs.
- Power BI executive dashboard.

These activities are assigned to Sprint 2 and Sprint 3 in our UC2 roadmap.

8. High-Level Solution

The proposed solution follows an enterprise data engineering approach:





The project repository follows the Bronze–Silver–Gold Medallion Architecture specified in the project structure.

9. Technology Stack

Area	Technology
ETL	Pentaho Data Integration (Spoon)
Database	PostgreSQL
Data Processing	Python / Pandas
Reporting	Power BI
Version Control	Git / GitHub
Documentation	Markdown / MS Word
Project Methodology	Agile Scrum

10. Sprint Plan

Sprint 0 – Project Initiation & Architecture

The purpose of Sprint 0 is to understand the healthcare business problem and design the enterprise data platform before development begins.

Key activities:

- Understand the healthcare business problem.
- Identify stakeholders.
- Identify healthcare data sources.
- Define project scope.
- Prepare project charter.
- Prepare sprint backlog.
- Design high-level architecture.
- Create Git repository.
- Create enterprise repository structure.

Sprint 1 – Data Discovery & Ingestion

The purpose of Sprint 1 is to build the healthcare raw-data landing zone by integrating healthcare data from multiple sources.

Key activities:

- Analyse Synthea source datasets.
- Prepare source inventory.
- Prepare data dictionary.
- Develop Pentaho ETL transformations.
- Ingest healthcare datasets.
- Validate incoming data.
- Load data into PostgreSQL staging tables.
- Implement logging and exception handling.
- Test ETL pipelines.
- Commit completed work to Git.

11. Expected Deliverables

Sprint 0

- Project Charter
- Sprint Backlog
- Business Requirement Document
- Git Repository
- High-level solution architecture

Sprint 1

- Source Inventory
 - Data Dictionary
 - Pentaho ETL Pipelines
 - PostgreSQL Staging Database
 - ETL testing evidence
 - Git Commit History
 - Technical documentation
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12. Success Criteria

The project/sprint will be considered successful when:

- Business requirements are addressed.
 - Pentaho ETL pipelines execute successfully.
 - Healthcare data is successfully loaded into PostgreSQL staging tables.
 - Data quality checks are documented.
 - Source-to-target mappings are maintained.
 - Code and ETL artifacts are committed to Git with meaningful commit messages.
 - Required documentation is completed.
 - Sprint review presentation is delivered.
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13. Key Risks and Mitigation

Risk	Impact	Mitigation
Different source file formats	ETL complexity	Use appropriate Pentaho input steps for each format
Missing or inconsistent data	Data quality issues	Implement validation and document data quality checks
Large healthcare datasets	ETL performance issues	Test pipelines with controlled batches and monitor execution
Incorrect source-to-target mapping	Incorrect data in PostgreSQL	Maintain a source-to-target mapping document

Risk	Impact	Mitigation
Configuration differences between team members	Deployment/execution issues	Maintain configuration and documentation in Git

14. Project Outcome

At the end of the project, the expected outcome is to:

- Analyse enterprise healthcare data requirements.
 - Build end-to-end ETL pipelines using Pentaho.
 - Integrate heterogeneous healthcare datasets.
 - Perform healthcare data profiling and quality assessment using Python.
 - Design and implement a PostgreSQL healthcare data warehouse.
 - Maintain metadata and data lineage.
 - Collaborate using Git and GitHub.
 - Develop executive healthcare dashboards.
 - Present an enterprise-grade healthcare data engineering solution using Agile sprint practices.
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15. Project Team Members

Name	Reg Number	Section
Smrithi Rajan	26MAS1002	MTech-AI and Data Science
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